Battery-Efficient Federated Learning: Research Blueprint

0. Validation — Is this a real research area?

Yes. This is an active subfield sometimes called **"Green FL"**, **"energy-aware FL"**, or **"sustainable FL"**, with published work through 2025–2026:

- **EAFL** (2022, ACM FedEdge workshop) — battery-aware client selection; reported accuracy gains up to 85% and drop-out reduction up to 2.45× vs. baselines by keeping low-battery clients from dying mid-round.
- **Eco-FL** (2024, *Computer Communications*) — energy-efficient client selection for sustainability.
- **Eecs-FL** (2025, *Springer J. Wireless Comm. & Networking*) — energy-efficient selection for AIoT.
- **FedZero** (2023, arXiv) — schedules training around renewable/excess energy availability, built on **Flower** + **Vessim**.
- **FedSynthesis** (2025, IEEE/ACM UCC) — Flower-based carbon-reduced FL framework.
- A dedicated **2025 survey**, *"Energy-Efficient Federated Learning at the Wireless Edge"* (Albert et al., *J. Electronic & Information Systems*), frames the whole space around three levers: **model compression, update sparsification, adaptive client scheduling**.
- **FedLuck** (2024) jointly tunes local-update frequency *and* gradient compression rate per device — closest existing work to a "joint optimization" angle.
- Real device-level energy profiling exists too: Flower's own paper deploys on Android phones and Nvidia Jetson boards to measure latency/energy footprint directly.

Conclusion: the topic is real, publishable, and has working code/tools behind it. The risk is *not* "does this exist" — it's "your exact framing (battery-aware client selection alone) has been done 3–4 times since 2021." So Section 2 below upgrades the idea into something with a cleaner novelty claim.

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1. Upgraded Problem Statement

Instead of just "battery-aware client selection," position the project as:

> **A joint, adaptive resource-controller for FL that decides, per device per round, whether to participate, how much local computation to do, and how much to compress the update — based on live battery/network state — while bounding the accuracy loss versus a non-energy-aware baseline.**

This reframes three separate known levers (selection / frequency / compression) as **one coordinated decision problem**, which is the genuinely under-explored angle (only FedLuck does 2 of the 3 jointly; nobody found combines all three with battery as first-class state).

Stretch upgrade (optional, higher novelty ceiling)

Add a **fairness/bias audit**: prove or disprove that always deprioritizing low-battery devices systematically excludes certain user/device segments (older phones, budget IoT sensors) from contributing data — and propose a correction (e.g., a "fairness credit" that occasionally forces inclusion of chronically low-battery clients). This is barely touched in the literature and gives the project an ethical/technical hook beyond pure efficiency.

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2. Research Questions

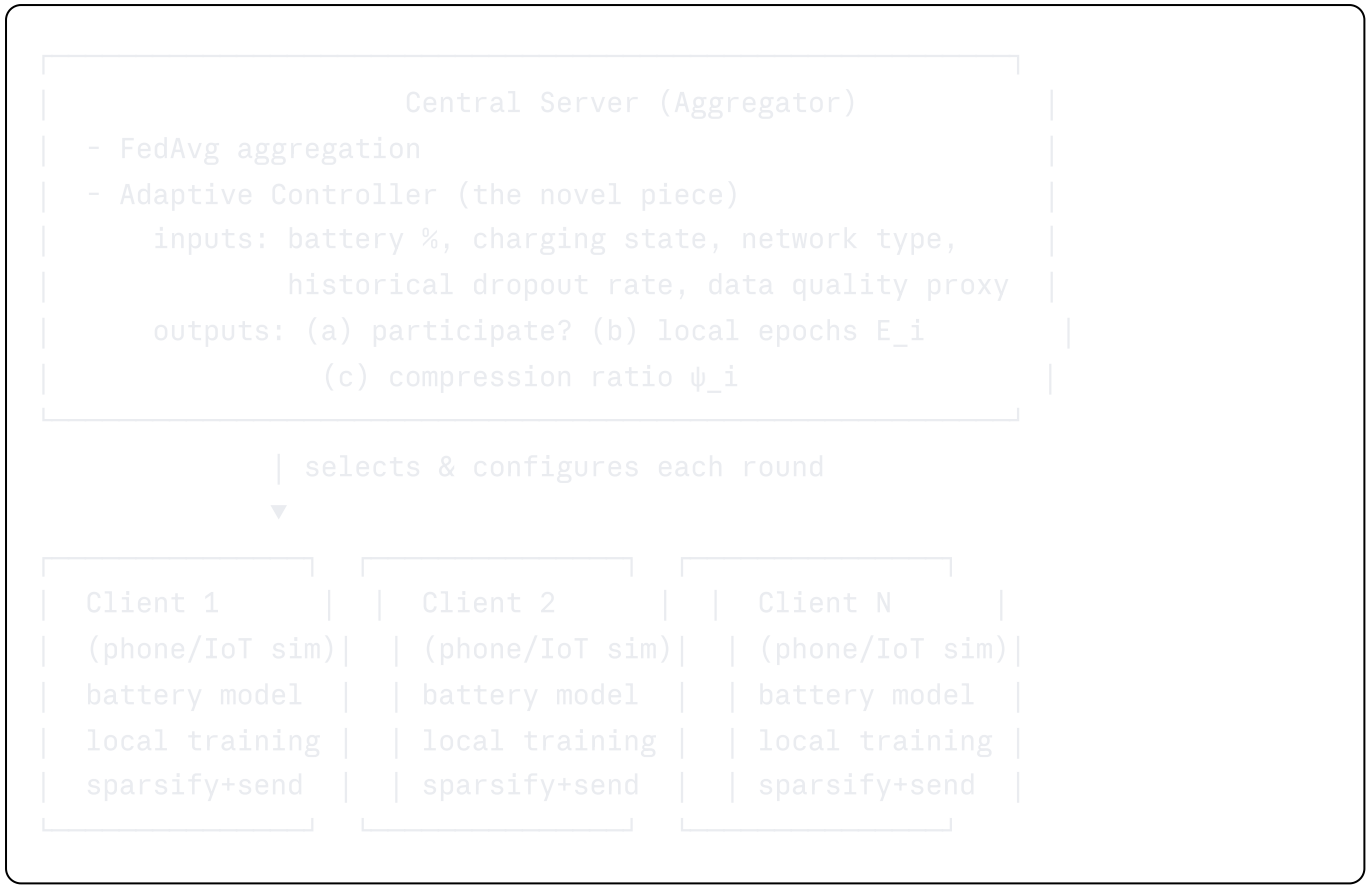
1. **RQ1 (Selection):** Does battery-and-network-aware client selection reduce aggregate energy consumption without degrading final model accuracy beyond an acceptable threshold (e.g., ⩽2% top-1 accuracy drop)?
2. **RQ2 (Frequency):** How does varying local epoch count / update frequency per device, conditioned on battery level, affect the energy–accuracy–convergence-speed trade-off?
3. **RQ3 (Compression):** At what gradient sparsification / quantization level does energy savings plateau while accuracy loss stays negligible (target: locate the "knee" of the curve, informally referenced as ~30% sparsity giving ~50% energy savings with negligible loss)?
4. **RQ4 (Joint):** Does a controller that jointly optimizes selection + frequency + compression outperform (Pareto-dominate) each lever applied independently?
5. **RQ5 (Fairness, stretch):** Does energy-aware scheduling introduce systematic exclusion bias across device/battery-capacity classes, and can a corrective mechanism mitigate it without erasing energy savings?

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3. Literature Map (what to read, why)

Paper	Lever	Key takeaway to build on
EAFL (2022)	Client selection	Battery-priority selection *improves* accuracy by reducing mid-training dropouts
Eco-FL (2024)	Client selection	Sustainability-focused selection metric design
Eecs-FL (2025)	Client selection	AIoT-specific constraints (mobility, heterogeneity)
FedZero (2023)	Scheduling	Uses Flower + Vessim for realistic energy-availability simulation
FedLuck (2024)	Frequency + compression	Jointly tunes local update frequency and compression rate — closest prior art to RQ4
Energy-Aware Gradient Pruning (2026 preprint)	Compression	Theoretical framework for battery-aware Top-K pruning
Green FL survey (2024, arXiv 2409.12626)	Compression	Explains sparsification/quantization trade-off mechanics in detail
Wireless Edge Energy survey (2025)	All three	Good overview chapter to cite in your intro/related work
Flower paper (2020)	Systems	How to actually measure energy on real Android/Jetson hardware

4. System Architecture



Controller logic (simplified policy you'll actually implement/train):

- Rule-based baseline: threshold on battery % (e.g., only train if >30%, reduce epochs if <50%, increase sparsification if <40%).
- Learned upgrade (if time allows): small RL agent (contextual bandit or lightweight Q-learning) that learns the mapping from device state → (participate, E_i, ψ_i) to maximize accuracy-per-joule.

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5. Tools & Frameworks (validated, currently maintained)

Purpose	Tool	Notes
FL orchestration	**Flower (flwr)**	Most widely used in this literature; supports real Android/embedded deployment later if you want physical validation
Energy/carbon simulation	**Vessim**	Used by FedZero; simulates realistic energy availability time series alongside Flower
On-device energy ground-truth (optional, stronger paper)	Android Battery Historian / `PowerTop` on Linux edge boards, or a physical USB power meter on a Raspberry Pi/Jetson Nano	Gives you *measured* joules instead of only estimated joules — big credibility boost
Energy estimation model (software-only path)	García-Martín et al. "Estimation of Energy Consumption in Machine Learning" (cited across this literature)	Formula-based estimate from FLOPs + hardware power draw, if you don't have physical hardware
Models/datasets	FEMNIST, CIFAR-10/100 (non-IID partitioned), Shakespeare (LEAF benchmark)	Standard in FL papers — makes your results comparable to prior work
Compression	Top-K gradient sparsification, quantization (8-bit/4-bit)	Well-documented baselines to implement from scratch or reuse from FedLuck's repo approach

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6. Experimental Design

6.1 Baselines to compare against

1. **Vanilla FedAvg** (no energy awareness) — upper bound on accuracy, upper bound on energy use.
2. **Random client selection** with fixed budget — common naive baseline.
3. **EAFL-style battery-priority selection only** — reproduces existing SOTA lever #1.
4. **Compression-only** (fixed Top-K sparsification, no selection logic) — reproduces lever #3.
5. **Your joint controller** — the contribution.
6. (Stretch) **Fairness-corrected joint controller.**

6.2 Metrics

- **Model quality:** top-1 accuracy / F1, communication rounds to convergence.
- **Energy:** estimated or measured joules per client per round; total system energy to reach target accuracy.
- **Fairness (stretch):** participation rate distribution across battery/device tiers; accuracy contribution variance across device classes.
- **System:** client dropout rate, wall-clock time to convergence.

6.3 Non-IID realism

Use Dirichlet-partitioned non-IID splits (standard practice) — energy-aware selection that also happens to always pick the same "healthy battery" clients can silently reduce data diversity, which is a real failure mode worth explicitly measuring (ties into RQ5).

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7. Suggested Timeline (for a semester/capstone-length project)

Phase	Weeks	Deliverable
1. Setup & reproduction	1–3	Flower + Vessim environment running; reproduce vanilla FedAvg on FEMNIST/CIFAR baseline
2. Implement levers independently	4–6	Battery-aware selection, adaptive local-epoch scheduling, gradient sparsification — each working standalone
3. Build joint controller	7–9	Rule-based joint policy; ablation vs. each lever alone
4. (Optional) learned controller	10–11	Contextual bandit/RL controller trained on simulated device traces
5. Fairness audit	11–12	Participation-bias analysis + mitigation
6. Real-hardware validation (optional but strong)	12–13	Deploy on 2–3 Android phones or Jetson Nano/Raspberry Pi with real power measurement
7. Write-up	13–14	Report/paper with Pareto curves: accuracy vs. energy for every method

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8. Expected Contribution / Success Criteria

- A clearly labeled **Pareto frontier plot** (accuracy vs. total energy) showing your joint controller dominates or matches each single-lever baseline.
- Quantified claim in the style of prior work, e.g., "X% energy reduction at ⩽2% accuracy cost" — this is the sentence reviewers/evaluators look for.
- If you do the fairness angle: a plot showing participation rate by device-battery-tier under naive vs. corrected policies.

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9. Risks & Mitigations

Risk	Mitigation
Simulated energy ≠ real energy, weak credibility	Even a small real-hardware validation (2–3 devices) massively strengthens the paper
Over-scoping (trying selection + frequency + compression + RL + fairness in one term)	Get rule-based joint controller working first; RL and fairness are stretch goals, not core deliverables
Non-IID + energy-aware selection interacting to reduce data diversity in confusing ways	Explicitly measure and report this rather than treating it as noise
Reviewers saying "just EAFL again"	Lead every write-up with the *joint* optimization framing and the fairness angle — that's your differentiator

10. Key References to Cite

- Beutel, D. J. et al. (2020). *Flower: A Friendly Federated Learning Framework.* arXiv:2007.14390.
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- García-Martín, E. et al. (2019). Estimation of Energy Consumption in Machine Learning. *J. Parallel & Distributed Computing*, 134.
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